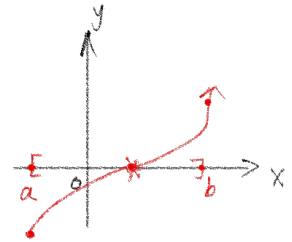


VIII Three hard theorems

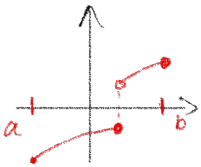
① Intermediate Value Theorem (IVT)

If f is continuous on $[a, b]$ and $f(a) < 0 < f(b)$,
then there exists some $c \in [a, b]$ s.t. $f(c) = 0$

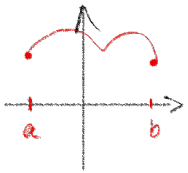


Checking that the assumptions are necessary

- If f were not continuous



- If $f(a), f(b) > 0$



Collary If f is cts. on $[a, b]$ and $f(a) < N < f(b)$

then $\exists c \in [a, b]$ s.t. $f(c) = N$.

PF. Define $g(x)$ on $[a, b]$ by $g(x) = f(x) - N$

- g is cts. b/c. $f(x)$ and $-N$ are cts.
- Want to show $g(a) < 0 < g(b)$

Since $g(a) = f(a) - N < 0$, $g(b) = f(b) - N > 0$

Apply IVT $\Rightarrow \exists c \in [a, b]$ s.t.

$$\begin{array}{c} g(c) = 0 \\ \parallel \\ f(c) = N \end{array} \Rightarrow f(c) = N$$

$$f(c) = N$$



Collary. If f cts on $[a, b]$ and $f(a) > 0 > f(b)$

Then $\exists c \in [a, b]$ s.t. $f(c) = 0$

These two collaries allow us to assume WOLOG that the cts. function f satisfies $f(a) < 0 < f(b)$.

Ex. Let $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ with n odd, then $f(x) = 0$ has a real root.

Pf. Idea. We know that

$$\lim_{x \rightarrow \infty} f(x) > 0$$

$$\lim_{x \rightarrow -\infty} f(x) < 0$$

So need some $a > b$ s.t. $f(a) > 0, f(b) < 0$

then apply IVT.

Pf. First note that f is cts:

We've shown that x is cts. (for any ϵ , take $\delta = \epsilon$).

and any constant func. is cts. (for any ϵ , take any δ).

And we've shown that sum & product of cts. function is cts. $\Rightarrow f$ is cts.

Now we want $[a, b]$ s.t. $f(a) > 0 > f(b)$.

Note $f(x) = x^n \left(1 + \frac{a_n}{x} + \frac{a_{n-1}}{x^2} + \dots + \frac{a_0}{x^n} \right)$ for $x \neq 0$

(As long as we choose $a, b \neq 0$, OK)

Goal. Show that $1 + \frac{a_n}{x} + \frac{a_{n-1}}{x^2} + \dots + \frac{a_0}{x^n}$ is a positive # very close to 1 for $|x|$ sufficiently large.

$$\left| \frac{a_{n-1}}{x} + \dots + \frac{a_0}{x^n} \right| \leq \left| \frac{a_{n-1}}{x} \right| + \dots + \left| \frac{a_0}{x^n} \right| \quad (*)$$

want

$$\leq \frac{1}{2}$$

$$\text{Then } -\frac{1}{2} \leq \frac{a_{n-1}}{x} + \dots + \frac{a_0}{x^n} \leq \frac{1}{2}$$

$$\text{So } \frac{1}{2} \leq 1 + \frac{a_{n-1}}{x} + \dots + \frac{a_0}{x^n} \leq \frac{3}{2}$$

We'll find x large, s.t.

$$\left| \frac{a_{n-i}}{x^i} \right| \leq \frac{1}{2n}, \quad 1 \leq i \leq n$$

Roughwork. if $i=1$, $\left| \frac{a_{n-1}}{x} \right| < \frac{1}{2n}$

need $x > 2n|a_{n-1}|$

$$i=2, \quad \left| \frac{a_{n-2}}{x^2} \right| < \left| \frac{a_{n-2}}{x} \right| < \frac{1}{2n}$$

need $x > 2n|a_{n-2}|$, etc.

$$\Rightarrow \text{want } x > \max\{2n|a_{n-1}|, 2n|a_{n-2}|, \dots, 2n|a_0|, 1\}.$$

We claim that $x > \max \{2n|a_{n-1}|, 2n|a_{n-2}|, \dots, 2n|a_0|, 1\}$ is good enough

call this M

Then RHS of $(*)$ becomes

$$\leq \left| \frac{a_{n-1}}{x} \right| + \dots + \left| \frac{a_0}{x^n} \right|$$

$$< \left| \frac{a_{n-1}}{x} \right| + \left| \frac{a_{n-2}}{x} \right| + \dots + \left| \frac{a_0}{x} \right|$$

$$< \frac{1}{2n} + \frac{1}{2n} + \dots + \frac{1}{2n} = \frac{1}{2}$$

$$\text{Then } \frac{1}{2} < \frac{a_{n-1}}{x} + \dots + \frac{a_0}{x^n} < \frac{1}{2}$$

$$\Rightarrow \frac{1}{2} < 1 + \frac{a_{n-1}}{x} + \dots + \frac{a_0}{x^n} < \frac{3}{2} \leftarrow \text{So for } x > M \text{ the stuff inside bracket is } > 0.$$

So if we look at the interval $[-2M, 2M]$

$$f(x) = x^n \left(1 + \frac{a_{n-1}}{x} + \dots + \frac{a_0}{x^n} \right) : \begin{cases} > 0, \text{ if } x = 2M \\ < 0, \text{ if } x = -2M \end{cases}$$

So apply IVT to f on $[-2M, 2M]$ and since $f(-2M) < 0 < f(2M)$ by IVT

$\exists c \in [-2M, 2M]$ s.t. $f(c) = 0$

i.e., c is a real root of f .



This ~~pf~~ actually gives a bound on the location of the root.

Ex. $f(x) = x^3 + 5x^2 - 10x + 20$

$$\text{Then } M = \max \{6 \cdot 5, 6 \cdot |-10|, 6 \cdot |20|, 1\} = 120$$

so f has a real root b/w. -120 and 120 .